Unit: Unit 7: Factoring
Topic: Expanding Special Patterns Perfect Square Trinomial

Name: ________________
Date: ____________

Quiz (Carolina Reaper)

Q1. A shipping company has x boxes to transport. If each truck can carry $(x+5)$ boxes, what is the expression for the total number of boxes transported by 2 trucks?

(A) $2x+10$
(B) $2x-10$
(C) x^2+10x
(D) x^2-10x
(E) $2x^2+10x$

Q2. The time it takes to travel from City A to City B is given by the expression $t=x^2-4x+4$, where t is time in hours and x is the speed in km/h. What type of trinomial is this expression?

(A) Perfect square trinomial
(B) Difference of cubes
(C) Sum of cubes
(D) Product of two binomials
(E) None of the above

Q3. The cost of shipping a package is given by the expression $C=(x+2)^2$, where x is the weight of the package in kg. What is the expanded form of this expression?

(A) x^2+2x+4
(B) x^2-2x-4
(C) x^2+4x+4
(D) x^2-4x+4
(E) x^2+4x-4

Q4. The distance between two cities is given by the expression $d=(x-3)^2$, where x is the speed of the vehicle in km/h. What is the value of x that makes $d=16$?

(A) $x=1$
(B) $x=3$
(C) $x=5$
(D) $x=7$
(E) $x=9$

Q5. The time it takes for a package to be delivered is given by the expression $t=x^2-9$, where t is time in days and x is the distance in km. What is the factored form of this expression?

(A) $(x-3)(x+3)$
(B) $(x-3)^2$
(C) $(x+3)^2$
(D) $(x-3)(x-3)$
(E) $(x+3)(x+3)$

Q6. A logistics company has x packages to deliver. If each delivery truck can carry $(x-2)$ packages, what is the expression for the total number of packages delivered by 3 trucks?

(A) $3x-6$
(B) $3x+6$
(C) x^2-6x
(D) x^2+6x
(E) $3x^2-6x$

Q7. The cost of fuel for a vehicle is given by the expression $C=(x+1)^2$, where x is the distance traveled in km. What is the expanded form of this expression?

(A) x^2+2x+1
(B) x^2-2x-1
(C) x^2+2x-1
(D) x^2-2x+1
(E) x^2+1

Q8. The time it takes for a vehicle to travel from City A to City B is given by the expression $t=x^2+4x+4$, where t is time in hours and x is the speed in km/h. What type of trinomial is this expression?

(A) Perfect square trinomial
(B) Difference of cubes
(C) Sum of cubes
(D) Product of two binomials
(E) None of the above

Open Ended Questions (FRQ)

Q9. A shipping company has a container that can hold x boxes. If each box has a weight of $(x+2)$ kg, what is the expression for the total weight of the boxes in the container? Show all work and simplify your answer.

Q10. The cost of delivering a package is given by the expression $C=x^2-4x+4$, where x is the distance traveled in km. Factor this expression and find the value of x that makes $C=0$. Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Ensure students have a clear understanding of perfect square trinomials and how to factor them.
- Tip 2: Encourage students to use real-world examples to illustrate their understanding of the concepts.
- Tip 3: Remind students to show all work and explain their reasoning for each problem.